

Bryce Richard

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Experience

Blissway

Denver, CO

Machine Learning Engineer

June 2025 - Present

- Founding member of the ML team at a Y Combinator-backed startup building AI-powered tolling and roadway safety infrastructure; designed and built core computer vision systems from the ground up.
- Redesigned the ML inference pipeline into **optimized microservices**, cutting inference costs by [X]% while improving throughput and reliability.
- Built a **vision-based vehicle triggering system** end to end — dataset curation, model training, and the production real-time triggering service — achieving [X]% precision and [X]% recall in the field.
- Developed **axle counting** and **lane classification** systems powering automated vehicle classification and toll assessment.
- Trained custom **license plate recognition** models to read plates and extract additional vehicle attributes for identification.

Carnegie Mellon University - Biorobotics Lab

Pittsburgh, PA

Graduate Research Assistant

August 2023 - May 2025

- Designed and implemented a **real-time sortation system** for a robotic disassembly pipeline, in collaboration with a major OEM, optimizing the recycling process for smart devices.
- Trained and fine-tuned **YOLOv8**, U-Net, and Mask-RCNN models for **semantic and instance segmentation**, achieving high precision in identifying and sorting components.
- Integrated a SCARA robotic arm with real-time vision-based decision-making using **ROS**; built a custom annotation tool with the **Segment Anything Model (SAM)** that cut labeling time by 50%.

Brigham Young University - FLOW Lab

Provo, UT

Research Assistant

August 2022 - May 2023

- Developed optimization tools that integrated bathymetry data to minimize the LCOE of offshore wind farms, decreasing the projected LCOE of projects by 10% after optimization.

Education

Carnegie Mellon University

Pittsburgh, PA

M.S. in Mechanical Engineering - Research | *QPA: 3.9/4.0*

May 2025

- Coursework: Advanced Computer Vision • Intermediate Deep Learning • Trustworthy AI • Systems & Toolchains for AI Engineers

Brigham Young University

Provo, UT

B.S. in Mechanical Engineering; Minor in Computer Science | *GPA: 3.9/4.0*

April 2023

Skills

Programming Languages: Python (advanced), C++

ML & Vision: PyTorch, OpenCV, YOLO, model optimization (ONNX/TensorRT), SAM, ROS

Infrastructure: Microservices, Docker, FastAPI, AWS (EC2, Lambda), Modal, MongoDB, PostgreSQL, Linux, CI/CD